

PAPER_1613 — U-238 Binding Energy per A = U-238 BE/A = $F \cdot K^5 + \beta^2 + \beta^3 + F \cdot \beta + 3 \approx 7.568 \text{ MeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Nuclear **Date:** June 18, 2026 **Status:** CLOSED — 0.033% residual

Observation

PAPER_1209II_S667 (Nuclear Unified Proof Set) gives the closed-form expression:

$$\text{U-238 BE/A} = F \cdot K^5 + \beta^2 + \beta^3 + F \cdot \beta + 3 \approx 7.568 \text{ MeV}$$

Residual to AME2020: **0.033%**.

UQFF Closed Identity

$$\text{U-238 BE/A} = F \cdot K^5 + \beta^2 + \beta^3 + F \cdot \beta + 3 \approx 7.568 \text{ MeV} \quad 0.033\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S667
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_u238_be_a_7_570"})`

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